Exercise J-M

1. Let $f: R \to R$ be a function such that $f(2) = 4$ and $f'(2) = 1$. Then, the value of $\lim\limits_{x\to 2}\dfrac{x^2f(2)-4f(x)}{x-2}$ is equal to : **[JEE(Main)July2021]**

माना $f: R \to R$ एक फलन है जिसके लिए $f(2) = 4$ तथा $f'(2) = 1$ है। तो $\lim\limits_{x\to 2}\dfrac{x^2f(2)-4f(x)}{x-2}$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 16 (B) 8 (C) 12 (D) 4

Ans. C

Sol. $\lim\limits_{x\to 2}\dfrac{x^2f(2)-4f(x)}{x-2}$ $\left(\dfrac{0}{0}\right)$

Use L'Hospital Rule

$\lim\limits_{x\to 2}\dfrac{2x\ f(2)-4f'(x)}{1}$

$\Rightarrow 4f(2) - 4f'(2) = 12$

2. Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\dfrac{1-x}{x}}\right)\right), 0 < x < 1$. Then: **[JEE(Main)Aug.2021]**

माना Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\dfrac{1-x}{x}}\right)\right), 0 < x < 1$ है, तो : **[JEE(Main)Aug.2021]**

(A) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$ (B) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$

(C) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$ (D) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$

Ans. B

Sol. Put $x = \sin^2\theta$

$f(x) = \cos\left(2\tan^{-1}\left(\sin\left(\cot^{-1}\sqrt{\dfrac{\cos^2\theta}{\sin^2\theta}}\right)\right)\right)$

$= \cos(2\tan^{-1}\sin\theta)$

let $\tan^{-1}(\sin\theta) = \theta$

$\tan\theta = \sin\theta$

$f = \cos(2\theta)$

$= \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$

$f(x) = \dfrac{1-x}{1+x}$

$f'(x) = \dfrac{-2}{(1+x)^2}$

$(1-x)^2 f'(x) = -2\left(\dfrac{1-x}{1+x}\right)^2$

$(1-x)^2 f'(x) = -2(f(x))^2$

3. If $y(x) = \cot^{-1}\left(\dfrac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right), x \in \left(\dfrac{\pi}{2}, \pi\right)$, then $\dfrac{dy}{dx}$ at $x = \dfrac{5\pi}{6}$ is : **[JEE(Main)Aug.2021]**

यदि $y(x) = \cot^{-1}\left(\dfrac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right), x \in \left(\dfrac{\pi}{2}, \pi\right)$ हैं, तो $\dfrac{dy}{dx}$ पर $x = \dfrac{5\pi}{6}$ का मान है :

[JEE(Main)Aug.2021]

(A) $\dfrac{1}{2}$ (B) $-\dfrac{1}{2}$ (C) –1 (D) 0

Ans. B

Sol. $y = \cot^{-1}\left(\dfrac{\left|\sin\dfrac{x}{2}+\cos\dfrac{x}{2}\right|+\left|\sin\dfrac{x}{2}-\cos\dfrac{x}{2}\right|}{\left|\sin\dfrac{x}{2}+\cos\dfrac{x}{2}\right|-\left|\sin\dfrac{x}{2}-\cos\dfrac{x}{2}\right|}\right)$

$y = \cot^{-1}\left(\dfrac{2\sin\dfrac{x}{2}}{2\cos\dfrac{x}{2}}\right)$

$y = \cot^{-1}\left(\tan\dfrac{x}{2}\right)$

$y = \cot^{-1}\left(\cot^{-1}\left(\dfrac{\pi}{2}-\dfrac{x}{2}\right)\right)$

$$y = \frac{\pi}{2} - \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{-1}{2}$$

4. If $y = y(x)$ is an implicit function of x such that $\log_e(x+y) = 4xy$, then $\frac{d^2y}{dx^2}$ at $x = 0$ is equal to _____.

[JEE(Main)Aug.2021]

यदि $y = y(x)$ का एक अस्पष्ट फलन इस प्रकार है कि $\log_e(x+y) = 4xy$ है, तो $x = 0$ पर $\frac{d^2y}{dx^2}$ बराबर है_____

[JEE(Main)Aug.2021]

Ans. 40

Sol. $l\,n(x+y) = 4xyx = 0 \quad \Rightarrow \quad y = 1$

$x + y = e^{4xy}$

$$1 + \frac{dy}{dx} = e^{4xy}\left(4y + 4x\frac{dy}{dx}\right) \quad ....(1)$$

$1 + y_1 = 1(4 + 0)$

$y_1 = 3$

Diff... (1)

$$\frac{d^2y}{dx^2} = e^{4yx}\left(4y + \frac{4xdy}{dx}\right)^2$$

$$+e^{4xy}\left(4\frac{dy}{dx} + 4\frac{dy}{dx} + 4x\frac{d^2y}{dx}\right)$$

Put $x = 0, y = 1, \frac{dy}{dx} = 3$

$y_2 = (4)^2 + (12 + \ 12 \ + 0)$

$y_2 = 40$

$$\left.\frac{d^2y}{dx^2}\right)_{x=0} = 40$$

5. If $y^{1/4} + y^{-1/4} = 2x$, and $(x^2-1)\frac{d^2y}{dx^2} + \alpha x\frac{dy}{dx} + \beta y = 0$, then $|\alpha-\beta|$ is equal to_____.

[JEE(Main)Aug.2021]

यदि $y^{1/4} + y^{-1/4} = 2x$ तथा $(x^2-1)\frac{d^2y}{dx^2} + \alpha x\frac{dy}{dx} + \beta y = 0$, है, तो $|\alpha-\beta|$ बराबर है_____

[JEE(Main)Aug.2021]

Ans. 17

Sol. $y^{\frac{1}{4}} + y^{-\frac{1}{4}} = 2x$

$$\left(\frac{1}{4}.y^{-\frac{3}{4}} - \frac{1}{4}y^{-\frac{5}{4}}\right)\frac{dy}{dx} = 2$$

$$\frac{1}{4y}\left(y^{\frac{1}{4}} - y^{-\frac{1}{4}}\right)\frac{dy}{dx} = 2$$

$$\sqrt{\left(y^{\frac{1}{4}} + y^{-\frac{1}{4}}\right)^2 - 4}\,\frac{dy}{dx} = 8y$$

$$\Rightarrow \sqrt{4x^2-4}\,\frac{dy}{dx} = 8y$$

$$\Rightarrow \sqrt{x^2-1}\,\frac{dy}{dx} = 4y$$

$$\Rightarrow \sqrt{x^2-1}\,\frac{d^2y}{dx^2} + \frac{1}{2\sqrt{x^2-1}}2x\frac{dy}{dx} = 4\frac{dy}{dx} = 4\frac{4y}{\sqrt{x^2-1}}$$

$$\Rightarrow (x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 16y = 0$$

$\alpha = 1, \beta = -16$

$|\alpha - \beta| = 17$

6. Let $f(x) = x^6 + 2x^4 + x^3 + 2x + 3$, $x \in R$. Then the natural number n for which $\lim\limits_{x\to 1}\frac{x^n f(1) - f(x)}{x-1} = 44$ is _____.

[JEE(Main)Aug.2021]

माना $f(x) = x^6 + 2x^4 + x^3 + 2x + 3$, $x \in R$ है। तब $\lim\limits_{x\to 1}\frac{x^n f(1) - f(x)}{x-1} = 44$ के लिए प्राकृतिक संख्या n है ___

[JEE(Main)Aug.2021]

Ans. 7

Sol. $f(x) = x^6 + 2x^4 + x^3 + 2x + 3, x \in R$

$f(1) = 9$

$$\lim_{x \to 1} \frac{9x^n - f(x)}{x - 1} = 44$$

Since limit on the left hand side is of the form $\left(\frac{0}{0}\right)$.

Applying L'Hospital Rule

$$\Rightarrow \lim_{x \to 1} \frac{9nx^{n-1} - f'(x)}{1} = 44$$

$$\Rightarrow 9n - f'(1) = 44 \quad .....(1)$$

$$\because f'(x) = 6x^5 + 8x^3 + 3x^2 + 2$$

$f'(1) = 19$

$\Rightarrow 9n - 19 = 44$

$9n = 63$

$n = 7 \cdot$

7. If $y = \tan^{-1}\left(\sec x^3 - \tan x^3\right), \frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, then : **[JEE(Main)June2022]**

यदि $y = \tan^{-1}\left(\sec x^3 - \tan x^3\right), \frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, है, तो : **[JEE(Main)June2022]**

(A) $xy'' + 2y' = 0$ (B) $x^2y'' - 6y + \frac{3\pi}{2} = 0$

(C) $x^2y'' - 6y + 3\pi = 0$ (D) $xy'' - 4y' = 0$

Ans. B

Sol. $$y = \tan^{-1}\left(\frac{1}{\cos x^3} - \frac{\sin x^3}{\cos x^3}\right)$$

$$y = \tan^{-1}\left(\frac{1 - \sin x^3}{\cos x^3}\right)$$

$$y = \tan^{-1}\left(\frac{1 - \cos\left(\frac{\pi}{2} - x^3\right)}{\sin\left(\frac{\pi}{2} - x^3\right)}\right)$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{x^3}{2}\right)\right)$$

$$-\frac{\pi}{2}<\frac{\pi}{4}-\frac{x^3}{2}<0$$

$$y=\frac{\pi}{4}-\frac{x^3}{2}$$

$$\frac{dy}{dx}=\frac{-3x^2}{2}$$

$$y'=\frac{-3}{2}x^2$$

$y'' = -3x$

Consider option (A) $xy'' + 2y' = 0$

$$x(-3x)+2\left(-\frac{3}{2}x^2\right)=0$$

which is wrong

(B) $x^2y''-6y+\frac{3\pi}{2}=0$

$$x^2(-3x)-6\left(\frac{\pi}{4}-\frac{x^3}{2}\right)+\frac{3\pi}{2}=0$$

which is correct option

8. Let $f: R \to R$ be definied as

$$f(x) = x^3 + x - 5$$

If $g(x)$ is a function such that $f(g(x)) = x,\ \forall\ 'x' \in R$, then $g'(63)$ is equal to _____.

[JEE(Main)June2022]

माना $f: R \to R$

$f(x) = x^3 + x - 5$

द्वारा परिभाषित है। यदि एक फलन $g(x)$ के लिए $f(g(x)) = x,\ \forall\ 'x' \in R$ है, तो $g'(63)$ बराबर है _____

[JEE(Main)June2022]

(A) $\frac{1}{49}$ (B) $\frac{3}{49}$ (C) $\frac{43}{49}$ (D) $\frac{91}{49}$

Ans. A

Sol. $f(x) = x^3 + x - 5$

$\Rightarrow f'(x) = 3x^2 + 1 \Rightarrow$ increasing function

$\Rightarrow$ invertible

$\Rightarrow g(x)$ is inverse of $f(x)$

$\Rightarrow g(f(x)) = x$

$\Rightarrow g'(f(x))f'(x) = 1$

$f(x) = 63$

$\Rightarrow x^3 + x - 5 = 63$
$\Rightarrow x = 4$
put $x = 4$

$$g'(f(4))f'(4) = 1$$
$$g'(63) \times 49 = 1 \qquad \{f'(4) = 49\}$$
$$g'(63) = \frac{1}{49}$$

9. Let f(x) be polynomial function such that $f(x) + f'(x) + f''(x) = x^5 + 64$. Then, the value of $\lim\limits_{x \to 1} \frac{f(x)}{x-1}$ is equal to: **[JEE(Main)June2022]**

माना f(x) एक बहुपद फलन है, जिसके लिए $f(x) + f'(x) + f''(x) = x^5 + 64$ है। तो $\lim\limits_{x \to 1} \frac{f(x)}{x-1}$ का मान बराबर है : **[JEE(Main)June2022]**

(A) –15 (B) –60 (C) 60 (D) 15

Ans. A

Sol. $\underset{x \to 1}{Lt} \frac{f(x)}{x-1} = f'(1) \left(\text{and } f(1) = 0\right)$

$f'(x) + f'(x) + t''(x) = x^5 + 64$

$f'(x) + f''(x) + f'''(x) = 5x^4$

$f''(x) + f'''(x) + f^{iv}(x) = 20x^3$

$f'''(x) + f^{iv}(x) + f^{v}(x) = 60x^2$

$\therefore f^{v}(x) f''(x) = 60x^2 - 20x^3$

$\Rightarrow 120 - f''(1) = 40 \Rightarrow f''(1) = 80$

Also $f(1) + f'(1) + f''(1) = 65 \Rightarrow f'(1) = -15$ Ans.

10. If $\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$, $|y| < 2$, then :

यदि $\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$, $|y| < 2$ हैं, तो :

(A) $x^2y'' + xy' - 25y = 0$ (B) $x^2y'' - xy' - 25y = 0$

(C) $x^2y'' - xy' + 25y = 0$ (D) $x^2y'' + xy' + 25y = 0$

Ans. D

Sol. $\cos^{-1}\left(\frac{y}{2}\right)=\log_e\left(\frac{x}{5}\right)^5$

$\frac{y}{2}=\cos\left(5\ell n\left(\frac{x}{5}\right)\right)$

$y'=-2\sin\left(5\ell n\left(\frac{x}{5}\right)\right)\times\frac{5}{x}$

$xy'=-10\sin\left(5\ell n\left(\frac{x}{5}\right)\right)$

diff. w.r.t. x

$xy''+y'=-10\cos\left(5\ell n\left(\frac{x}{5}\right)\right)\times\frac{5}{x}$

$x^2y''+xy'=-50\cos\left(5\ell n\frac{x}{5}\right)$

$x^2y''+xy'=-25y$

$x^2y''+xy'+25y=0$

11. Let $f(x)=\begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a \in R$. Then the sum of squares of all the values of a, for which $2f'(10)-f'(5)+100=0$, is : **[JEE(Main)June2022]**

माना $f(x)=\begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a \in R$ है। तो a के सभी मानों, जिनके लिए $2f'(10)-f'(5)+100=0$ है, के वर्गों का योगफल है : **[JEE(Main)June2022]**

(A) 117 (B) 106 (C) 125 (D) 136

Ans. C

Sol. $f(x)=a(a^2+ax)+1(a^2x+ax^2)$

$f(x)=a^3+2a^2x+ax^2$

$f'(x)=2a^2+2ax$

$2f'(10)-f'(5)+100=0$

$2\left[2a^2+20a\right]-\left(2a^2+10a\right)+100=0$

$2a^2+30a+100=0$

$a^2+15a+50=0$

$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$= 225 - 100 = 125$

12. Let $f : R \to R$ safisfy $f(x+y) = 2^x f(y) + 4^y f(x), \forall x, y \in R$. If $f(2) = 3$ then $14.\dfrac{f'(4)}{f'(2)}$ is equal to ___. **[JEE(Main)June2022]**

माना $f : R \to R$, $f(x+y) = 2^x f(y) + 4^y f(x), \forall x, y \in R$ को संतुष्ट करता है। यदि $f(2) = 3$ है, तो $14.\dfrac{f'(4)}{f'(2)}$ बराबर है ___. **[JEE(Main)June2022]**

Ans. 248

Sol. $f(x+y) = 2^x f(y) + 4^y f(x)$

$f'(x+y)\times 1 = 2^x \ell n 2 f(y) + 4^y f'(x)$

$x = y = 2$

$f'(4) = 4\ell n 2 f(2) + 16 f'(2)$

$f'(4) - 16 f'(2) = 12\ell n 2$ ———(1)

$f'(x+y)\times 1 = 2^x f'(y) + f(x) 4^y \ell n 4$

$x = y = 2$

$f'(4) = 4f'(2) + 48\ell n 4$

$f'(4) - 4f'(2) = 48\ell n 4$ ———(2)

$12 f'(2) = 48\ell n 4 - 12\ell n 2$

$12 f'(2) = 96\ell n 2 - 12\ell n 2$

$f'(2) = 7\ell n 2$

$f'(4) - 16\times 7\ell n 2 = 12\ell n 2$

$f'(4) = 124\ell n 2$

$\dfrac{f'(4)}{f'(4)} = 124\times 2 = 248$

13. If $y(x) = (x^x)^x$, $x > 0$, then $\dfrac{d^2x}{dy^2} + 20$ at $x = 1$ is equal to _____. **[JEE(Main)June2022]**

यदि $y(x) = (x^x)^x$, $x > 0$ है, तो $x = 1$ पर $\dfrac{d^2x}{dy^2} + 20$ का मान बराबर है _____. **[JEE(Main)June2022]**

Ans. 16

Sol. $y = x^{x^2}$ at $x = 1, y = 1$

$\ell n\, y = x^2 \ell n\, x$

$$\frac{1}{y}\frac{dy}{dx} = 2x\ell nx + x^2.\frac{1}{x}$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{y\left(2x\,\ell n\,x + x\right)}$$

Diff wrt y

$$\frac{d^2x}{dy^2} = \frac{-1}{y^2\left(2x\,\ell n\,x + x\right)^2}.\left[\left(2x\,\ell n\,x + x\right) + y\left(2x.\frac{1}{x} + 2\ell n\,x + 1\right)\frac{dx}{dy}\right]$$

put x = 1, y = 1

$$\frac{d^2x}{dy^2} = \frac{-1}{(1)(0+1)^2}\left[(0+1) + (1)(2+0+1).\frac{1}{(1)(0+1)}\right]$$

$= -(1 + 3) = -4$

$$\Rightarrow \frac{d^2x}{dy^2} + 20 = -4 + 20 = 16$$

14. The value of $\log_e 2\frac{d}{dx}\left(\log_{\cos x}\cos ec\,x\right)$ at $x = \frac{\pi}{4}$ is : **[JEE(Main)July2022]**

$x = \frac{\pi}{4}$ पर $\log_e 2\frac{d}{dx}\left(\log_{\cos x}\cos ec\,x\right)$ का मान है : **[JEE(Main)July2022]**

(A) $-2\sqrt{2}$ (B) $2\sqrt{2}$ (C) -4 (D) 4

Ans. D

Sol. $\log_e 2\frac{d}{dx}\left(\log_{\cos x}\cos ecx\right)$

Let,

$y = \log_{\cos x}\text{cosec}x$

$$y = -\frac{\ln\left(\sin x\right)}{\ln\left(\cos x\right)}$$

$$\frac{dy}{dx} = -\frac{\left[\cot x.\ln\left(\cos x\right) + \tan x.\ln\left(\sin x\right)\right]}{\left(\ln\left(\cos x\right)\right)^2}$$

$$\left.\frac{dy}{dx}\right)_{x=\frac{\pi}{4}} = \frac{4}{\ln 2}$$

Now,

$$\Rightarrow \log_e 2.\frac{4}{\ln 2} = 4$$

15. Let $x(t)=2\sqrt{2}\cot t\sqrt{\sin 2t}$ and $y(t)=2\sqrt{2}\sin t\sqrt{\sin 2t}$, $t\in\left(0,\frac{\pi}{2}\right)$. Then $\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$ at $t=\frac{\pi}{4}$ is equal to :

[JEE(Main)July2022]

माना $x(t)=2\sqrt{2}\cot t\sqrt{\sin 2t}$ तथा $y(t)=2\sqrt{2}\sin t\sqrt{\sin 2t}$, $t\in\left(0,\frac{\pi}{2}\right)$ हैं। तो $\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$ पर $t=\frac{\pi}{4}$ बराबर है :

[JEE(Main)July2022]

(A) $\frac{-2\sqrt{2}}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{-2}{3}$

Ans. D

Sol. $x=2\sqrt{2}\cos t\sqrt{\sin 2t}$

$$\frac{dx}{dt}=\frac{2\sqrt{2}\cos 3t}{\sqrt{\sin 2t}}$$

$$y(t)=2\sqrt{2}\sin t\sqrt{\sin 2t}$$

$$\frac{dy}{dt}=\frac{2\sqrt{2}\sin 3t}{\sqrt{\sin 2t}}$$

$$\frac{dy}{dx}=\tan 3t$$

$$\frac{dy}{dx}=-1 \text{ at } t=\frac{\pi}{4}$$

$$\frac{d^2y}{dx^2}=\frac{3}{2\sqrt{2}}\sec^3 3t.\sqrt{\sin 2t}=-3 \text{ at } t=\frac{\pi}{4}$$

$$\therefore \frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}=\frac{1+1}{-3}=-\frac{2}{3}$$

16. For the curve C : $(x^2+y^2-3)+(x^2-y^2-1)^5=0$, the value of $3y'-y^3y''$, at the point (α,α), $\alpha>0$, on C, is equal to _____.

[JEE(Main)July2022]

वक्र C : $(x^2+y^2-3)+(x^2-y^2-1)^5=0$ के बिन्दु (α,α), $\alpha>0$ पर $3y'-y^3y''$ का मान बराबर है _____

[JEE(Main)July2022]

Ans. 16

Sol. (α, α) lies on

$C: x^2 + y^2 - 3 + x^2 - y^2 - 1^5 = 0$

Put $(\alpha, \alpha), 2\alpha^2 - 3 + -1^5 = 0$

$\Rightarrow \alpha = \sqrt{2}$

Now, differentiate C

$2x + 2y.y' + 5\left(x^2 - y^2 - 1\right)^4 \left(2x - 2yy'\right) =$ ....(1)

At $\left(\sqrt{2}, \sqrt{2}\right)$

$\sqrt{2} + \sqrt{2}y' + 5(-1)^4 \left(\sqrt{2} - \sqrt{2}y'\right) = 0$

$\Rightarrow y' = \frac{3}{2}$ .....(2)

Diff. (1) w.r.t. x

Again, Diff. (1) w.r.t. x

$1 + (y')^2 + yy'' + 20\left(x^2 - y^2 - 1\right)^3 (x - yy')^2 .2$

$+5\left(x^2 - y^2 - 1\right)^4 \left(1 - (y')^2 - yy''\right) = 0$

At $\left(\sqrt{2}, \sqrt{2}\right)$ and $y' = \frac{3}{2}$

We have,

$\left(1 + \frac{9}{4}\right) + \sqrt{2}y'' - 40\left(\sqrt{2} - \sqrt{2}.\frac{3}{2}\right)^2$

$+5(1)\left(1 - \frac{9}{4} - \sqrt{2}y''\right) = 0$

$\Rightarrow 4\sqrt{2}y'' = -23$

$\therefore 3y' - y^3 y'' = \frac{9}{2} + \frac{23}{2} = 16$